Chapter 15 — Basic Separation Techniques

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Filtration and crystallization

Q.1 Explain filtration and crystallization and the choice of solvent for crystallization.

Filtration separates an insoluble solid from a liquid. The solid left on the filter paper is the residue, and the liquid that passes through is the filtrate. It is useful for mixtures such as sand and water.

Crystallization separates a soluble solid from its solution. A suitable solvent dissolves the solute well when hot but poorly when cold, does not react with the solute, and allows impurities to remain in solution or be removed by hot filtration. Slow cooling gives larger and purer crystals because the crystal lattice forms more selectively.

Important Conceptual Short Questions — Topic 1

1. Why is filtration unsuitable for separating salt from water?

Salt dissolves in water and passes through the filter paper with the water. Filtration only removes insoluble solids.

2. How does slow cooling improve crystal purity?

Slow cooling allows particles to arrange into an ordered lattice and excludes many impurities, giving larger and purer crystals.

3. Why must a crystallization solvent dissolve the solute only sparingly when cold?

If the solute remains highly soluble when cold, it will stay in solution and few crystals will form on cooling.

4. How can insoluble impurities be removed during crystallization?

The hot solution is filtered before cooling, so insoluble impurities are left on the filter paper while the dissolved solute passes through.

Topic 2 — Distillation and fractional distillation

Q.2 Describe simple distillation and fractional distillation.

Simple distillation separates a liquid from a dissolved solid or from liquids with very different boiling points. The mixture is heated, the more volatile component vaporizes, and the vapour is condensed and collected as distillate.

Fractional distillation separates miscible liquids with close boiling points. A fractionating column provides repeated vaporization and condensation, so the vapour becoming richer in the more volatile component rises to the top. It is used

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in petroleum refining and ethanol-water separation.

Important Conceptual Short Questions — Topic 2

1. Why does fractional distillation separate liquids with close boiling points?

Repeated condensation and vaporization in the column enrich the vapour in the more volatile liquid, improving separation beyond a single distillation.

2. How does a fractionating column improve separation?

It provides a large surface area and a temperature gradient, allowing many equilibrium stages between rising vapour and descending liquid.

3. Why is the thermometer bulb placed at the side-arm of the distillation flask?

It measures the temperature of the vapour entering the condenser, which indicates the boiling point of the component being collected.

4. How can distillation be used to obtain pure water from salt water?

Water vaporizes on heating while salts remain behind. The vapour is condensed to give pure water.

Topic 3 — Chromatography and Rf value

Q.3 Explain paper chromatography and the use of the retardation factor.

Chromatography separates components of a mixture based on their different distribution between a stationary phase and a mobile phase. In paper chromatography, the stationary phase is water held in the paper and the mobile phase is a solvent moving by capillary action.

Components that are more soluble in the mobile phase and less strongly attracted to the stationary phase move farther. The retardation factor is R_f = distance moved by solute / distance moved by solvent front. Under the same conditions, R_f values help identify substances.

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Important Conceptual Short Questions — Topic 3

1. Why do different components move different distances in chromatography?

They have different solubilities in the mobile phase and different attractions to the stationary phase, so they travel at different rates.

2. How can R_f values identify an unknown substance?

If the unknown and a known substance have the same R_f under identical conditions, they may be the same compound.

3. Why must the solvent front be marked immediately after the run?

The solvent evaporates quickly, and the front position is needed to calculate R_f accurately.

4. How are colourless substances detected on a chromatogram?

They are located using UV light, iodine vapour or suitable locating agents that produce coloured spots.

Topic 4 — Checking purity of a product

Q.4 How can the purity of a separated product be checked?

A pure solid has a sharp melting point, while an impure solid usually melts over a lower and broader temperature range. A pure liquid has a constant boiling point at a given pressure. Chromatography can also check purity: a pure substance gives one spot under suitable conditions.

These tests are important because separation methods may leave traces of solvent, starting material or by-products. Melting point, boiling point and chromatography together give reliable evidence of purity.

Important Conceptual Short Questions — Topic 4

1. Why does an impure solid melt over a range of temperature?

Impurities disrupt the crystal lattice and weaken intermolecular forces, so different parts melt at different temperatures.

2. How can chromatography show that a product is impure?

More than one spot indicates that more than one component is present.

3. Why is a sharp melting point evidence of purity?

A pure crystalline substance has a uniform lattice and melts at a definite temperature.

4. How can boiling point be used to check liquid purity?

A pure liquid boils at a constant temperature at a given pressure, while impurities often cause a boiling range or change the boiling point.

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